

Stiefel and Grassmann Manifolds: A Geometric Foundation for Orthogonality and Subspace Problems in Machine Learning

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Abstract

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1 Introduction

2 Preliminaries

We follow the conventions of [2] and [1].

2.1 Basic Definitions

For the definitions of smooth manifolds and smooth maps, we follow Segal [2].

Definition 2.1. Let X be a topological space. A **Co-ordinate Chart** on X is the pair (U, φ) where $U \subset X$ is open, and φ is a homeomorphism

$$\varphi : U \xrightarrow{\sim} \varphi(U)$$

where $\varphi(U)$ is an open subset $\varphi(U) \subset \mathbb{R}^n$, for some nonnegative integer n .

Definition 2.2. Let X be a topological space, and fix a nonnegative integer n . X is an **n-dimensional topological manifold** if and only if X is Hausdorff and second countable, and for any point $x \in X$, there exists a co-ordinate chart,

$$\varphi : U \xrightarrow{\sim} \varphi(U) \subset \mathbb{R}^n$$

where $x \in U$, and $\varphi(U)$ is open in $\mathbb{R}^n$.

This is to say, that X is locally Euclidean, i.e. every point of X has an open neighbourhood homeomorphic to an open subset of $\mathbb{R}^n$.

We require that X is Hausdorff, and second countable, as to exclude certain pathological examples.

Definition 2.3. Let X be a topological manifold, and let (U, φ_1) and (V, φ_2) be two co-ordinate charts on X . The **transition function** between these two co-ordinate charts is the function:

$$\phi_{21} = \varphi_2 \circ \varphi_1^{-1} : \varphi_1(U \cap V) \xrightarrow{\sim} \varphi_2(U \cap V)$$

Notice, it is automatically a homeomorphism, as it is the composition of two homeomorphisms.

Its inverse is the transition function in the opposite direction:

$$\phi_{21}^{-1} = \phi_{12} = \varphi_1 \circ \varphi_2^{-1} : \varphi_2(U \cap V) \xrightarrow{\sim} \varphi_1(U \cap V)$$

Definition 2.4. Let X be a topological manifold. An **atlas** on X is a collection of co-ordinate charts on X :

$$\varphi_i : U_i \xrightarrow{\sim} \tilde{U}_i \subset \mathbb{R}^n$$

indexed by some, possibly infinite, set I , such that:

$$\bigcup_{i \in I} U_i = X$$

An atlas is a **smooth atlas** if and only if for any two co-ordinate charts in the atlas, the transition function

$$\phi_{ji} = \varphi_j \circ \varphi_i^{-1} : \varphi_i(U_i \cap U_j) \xrightarrow{\sim} \varphi_j(U_i \cap U_j), \quad i, j \in I$$

is a smooth function, i.e. it is infinitely differentiable.

Definition 2.5. Let X be a topological manifold, and let

$$\mathcal{A} = \{(U_i, \varphi_i); i \in I\}$$

be a smooth atlas on X . Let (U, φ) be any coordinate chart on X . The chart (U, φ) is **compatible** with the atlas $\mathcal{A}$ if and only if the transition functions between (U, φ) and any chart in $\mathcal{A}$ are diffeomorphisms, i.e. they are smooth functions with smooth inverses.

In other words, the new chart (U, φ) is compatible with the atlas $\mathcal{A}$ if and only if the union $\mathcal{A} \cup \{(U, \varphi)\}$ is still a smooth atlas.

Let $\mathcal{A}, \mathcal{B}$ be smooth atlases on X . The atlases $\mathcal{A}$ and $\mathcal{B}$ are **compatible** if and only if every chart in $\mathcal{A}$ is compatible with $\mathcal{B}$, and vice versa.

Corollary 2.6 ([2, Corollary 2.20]). *Compatibility is an equivalence relation on smooth atlases.*

Definition 2.7. A **smooth manifold** is a topological manifold X together with an equivalence class $[\mathcal{A}]$ of compatible smooth atlases on X .

The equivalence class $[\mathcal{A}]$ of compatible smooth atlases is called a **smooth structure** on X .

Definition 2.8. Let X, Y be smooth manifolds, of dimension m and n respectively, and let

$$H : X \longrightarrow Y$$

be a continuous function. Fix a point $x \in X$. H is **smooth at x** if and only if there exists a chart (U, φ) compatible with the smooth structure on X , with $x \in U$, and a chart (V, ψ) compatible with the smooth structure on Y , with $H(U) \subset V$, such that the composition

$$\psi \circ H \circ \varphi^{-1} : \varphi(U) \longrightarrow \psi(V)$$

is smooth at $\varphi(x)$.

The function H is **smooth** if and only if it is smooth at every point $x \in X$.

Definition 2.9. Let X, Y be smooth manifolds. A function

$$H : X \longrightarrow Y$$

is a **diffeomorphism** if and only if H is a bijection, and both H and H^{-1} are smooth.

If there exists a diffeomorphism between X and Y , they are **diffeomorphic**.

Definition 2.10. Following Segal, let X be a set. A **pseudo-chart** on X is the pair (U, φ) , where $U \subset X$ and

$$\varphi : U \longrightarrow \varphi(U)$$

is a bijection onto an open subset $\varphi(U) \subset \mathbb{R}^n$.

A collection $\mathcal{A} = \{(U_i, \varphi_i) : i \in I\}$ of pseudo-charts, with the same dimension n , is called a pseudo-atlas on X if

$$\bigcup_{i \in I} U_i = X.$$

Proposition 2.11 ([2, Proposition 2.26]). *Let X be a set, and let $\mathcal{A} = \{(U_i, \varphi_i) : i \in I\}$ be a pseudo-atlas for X . Suppose that for any pair $(U_1, \varphi_1), (U_2, \varphi_2)$ of pseudo-charts in $\mathcal{A}$, we have that:*

1. *both $\varphi_1(U_1 \cap U_2) \subset \varphi_1(U_1)$ and $\varphi_2(U_1 \cap U_2) \subset \varphi_2(U_2)$ are open subsets, and*
2. *the transition function is continuous.*

Then there exists a unique topology on X such that each pseudo-chart (U_i, φ_i) is a coordinate chart on X .

Corollary 2.12 ([2, Corollary 2.28]). *If we further assume that each transition function ϕ_{21} is smooth, then $[\mathcal{A}]$ is a smooth structure on the topological manifold X , provided the resulting topology is Hausdorff and second-countable.*

For embedded submanifolds and their tangent spaces, we follow the conventions of Boumal [1].

Definition 2.13. Let $\mathcal{E}$ be a real vector space of dimension d . A non-empty subset $\mathcal{M} \subset \mathcal{E}$ is a **smooth embedded submanifold** of $\mathcal{E}$ of dimension n , if either

1. $n = d$ and $\mathcal{M}$ is open in $\mathcal{E}$, also denoted as an **open submanifold**, or
2. $n = d - k$ for some $k \geq 1$ and, for each $x \in \mathcal{M}$, there exists an open neighbourhood $U \subset \mathcal{E}$ of x and a smooth function

$$h : U \longrightarrow \mathbb{R}^k$$

such that

- (a) If y is in U , then $h(y) = 0$ if and only if $y \in \mathcal{M}$; and
- (b) $\text{rank } Dh(x) = k$, where $Dh(x) : \mathcal{E} \rightarrow \mathbb{R}^k$ is the derivative of h at x .

Such a function h is called a **local defining function** for $\mathcal{M}$ at x .

Definition 2.14. Let $\mathcal{M}$ be a smooth embedded submanifold of a finite-dimensional real vector space $\mathcal{E}$. The **tangent space** to $\mathcal{M}$ at $x \in \mathcal{M}$ is the set

$$T_x \mathcal{M} = \{c'(0) \mid c : I \longrightarrow \mathcal{M} \text{ is smooth and } c(0) = x\},$$

where I is any open interval containing 0. We regard c as a curve in $\mathcal{E}$ when discussing its smoothness and derivatives.

That is, $v \in T_x \mathcal{M}$ if and only if there exists a smooth curve on $\mathcal{M}$ passing through x at $t = 0$ with velocity v .

Vectors in $T_x \mathcal{M}$ are called **tangent vectors** to $\mathcal{M}$ at x .

By [1, Theorem 3.15], $T_x \mathcal{M}$ is a subspace of $\mathcal{E}$, and

$$\dim T_x \mathcal{M} = \dim \mathcal{M},$$

independently of x .

2.2 Examples

Example 2.15.

Example 2.16.

3 Motivation

4 The Stiefel Manifold

Definition 4.1.

Remark 4.2.

5 The Grassmann Manifold

Lemma 5.1.

Proof.

□

Proposition 5.2.

Proof.

□

6 From Stiefel to Grassmann

Theorem 6.1.

Proof.

□

Corollary 6.2.

Proof.

□

7 Applications

7.1 $\text{Gr}(2, 4)$ in full

7.2 PCA on the Grassmannian

8 Discussion

9 Further Directions

10 Conclusion

References

- [1] Nicolas Boumal, *An introduction to optimization on smooth manifolds*, Cambridge University Press, 2023.
- [2] Ed Segal, *Manifolds*, 2016, Lecture notes, University College London.